

NLO Equations

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Abstract

1 Electric Dipole

$$\mu = \sqrt{\mu_x^2 + \mu_y^2 + \mu_z^2}$$

2 Polarizability

$$\alpha = \frac{xx + yy + zz}{3}$$

3 First Hyperpolarizability

The tensor components β_i generated by Gaussian lack multiplication by a factor of $\frac{1}{3}$. Additionally, the conversion from the Taylor series convention to the perturbational series convention involves multiplying by a $\frac{1}{2}$ factor, when considering the 3 possible beta tensors founded in Gaussian 16.0. This results in a total factor of $\frac{1}{6}$ to obtain β in the latter convention, as illustrated below:

$$\beta_i = \beta_{iii} + \frac{1}{3} \sum_{i \neq j} (\beta_{ijj} + \beta_{jij} + \beta_{jji})$$
$$\beta_i^{G16} = 3\beta_i$$

$$\beta^B = \frac{1}{2} \beta^T$$
$$\beta^B = \frac{1}{6} \beta^{G16}$$

3.1 T convention

$$\beta = \sqrt{\left(\frac{x}{3}\right)^2 + \left(\frac{y}{3}\right)^2 + \left(\frac{z}{3}\right)^2}$$

3.2 B convention

$$\beta = \sqrt{\left(\frac{x}{6}\right)^2 + \left(\frac{y}{6}\right)^2 + \left(\frac{z}{6}\right)^2}$$

4 Second Hyperpolarizability

The tensor components generated by Gaussian lack multiplication by a factor of $\frac{1}{6}$. Additionally, the conversion from the Taylor series convention to the perturbational series convention involves multiplying by a $\frac{1}{6}$ factor, when considering the 3 possible gamma tensors founded in Gaussian 16.0. This results in a total factor of $\frac{1}{36}$ to obtain γ in the latter convention, as illustrated below:

$$\begin{aligned}\frac{3}{2}\gamma^B &= \frac{1}{4}\gamma^T \\ \gamma^T &= \frac{1}{6}\gamma^{G16} \\ \gamma^B &= \frac{1}{36}\gamma^{G16}\end{aligned}$$

γ represents the overall gamma property calculated as discussed in the following sections.

4.1 Gamma Static (0;0,0,0)

4.1.1 T convention

$$\gamma = \frac{1}{5} \left(\frac{xxxx}{6} + \frac{yyyy}{6} + \frac{zzzz}{6} + 2\frac{xyxy}{6} + 2\frac{xxzz}{6} + 2\frac{yyzz}{6} \right)$$

4.1.2 B convention

$$\gamma = \frac{1}{5} \left(\frac{xxxx}{36} + \frac{yyyy}{36} + \frac{zzzz}{36} + 2\frac{xyxy}{36} + 2\frac{xxzz}{36} + 2\frac{yyzz}{36} \right)$$

4.2 Gamma Kerr effect (-w;w,0,0)

4.2.1 T convention

$$\gamma = \frac{1}{5} \left(\frac{xxxx}{6} + \frac{yyyy}{6} + \frac{zzzz}{6} + \frac{xxyy}{6} + \frac{yyxx}{6} + \frac{xxzz}{6} + \frac{zzxx}{6} + \frac{yyzz}{6} + \frac{zzyy}{6} \right)$$

4.2.2 B convention

$$\gamma = \frac{1}{5} \left(\frac{xxxx}{36} + \frac{yyyy}{36} + \frac{zzzz}{36} + \frac{xxyy}{36} + \frac{yyxx}{36} + \frac{xxzz}{36} + \frac{zzxx}{36} + \frac{yyzz}{36} + \frac{zzyy}{36} \right)$$

4.3 Gamma EFISH (-2w,w,w,0)

4.3.1 T convention

$$\gamma = 3 \left(\frac{xxxx}{6} + \frac{yyyy}{6} + \frac{zzzz}{6} \right) + 2 \left(\frac{xyxy}{6} + \frac{xzxz}{6} + \frac{yzyz}{6} + \frac{yxyx}{6} + \frac{zxzx}{6} + \frac{zyzy}{6} \right) \\ + \frac{xyyx}{6} + \frac{xzzx}{6} + \frac{yxyx}{6} + \frac{yzzz}{6} + \frac{zxxz}{6} + \frac{zyyz}{6}$$